
Measurements & Instruments

Exercise Book

IESE Exercise Book 1 of 3

Problems only — attempt them first; answers (final results, no worked steps) are at the back.

How to use this book

This is the problem set distilled from the *Measurements & Instruments* solving manual. It contains **only exercises and their answers** — no theory and no worked derivations. Each chapter mirrors a chapter of the manual, so when you are stuck on a *method*, open the manual at the same chapter and read its Protocol and Worked Example.

Key Idea

Work each exercise to a number *with its unit and uncertainty* before turning to Chapter 7. The answer key gives only the final result, so it confirms *what* you should get, never *how* — that is the point of working it yourself.

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Chapter 1

Measurements and the SI system

Exercise 1.1 Direct or indirect?

Classify each as a *direct* or *indirect* method, and name the quantities whose uncertainty enters the result: (a) timing 10 pendulum swings with a stopwatch to get one period; (b) measuring a temperature with a calibrated thermometer; (c) finding the input resistance of an amplifier from the divider $V_M = V_S R_{in}/(R + R_{in})$.

Exercise 1.2 Same resistor, two methods

A 50.0Ω resistor is read both on a DMM set to “ Ω ” and by computing $R = V/I$ from separate V and I readings. (a) Which method is direct, which indirect? (b) For each, state what makes the result traceable or what uncertainties combine.

Chapter 2

Measurements and uncertainty

Exercise 2.1 The four basic rules

Take $a = (15 \pm 0.5) \text{ cm}$ and $b = (10 \pm 0.5) \text{ cm}$. Give the value and the relative uncertainty of (a) the sum $a + b$; (b) the difference $a - b$; (c) the product of two $(10 \pm 0.5) \text{ cm}$ lengths; (d) the ratio of two $(10 \pm 0.5) \text{ cm}$ lengths.

Exercise 2.2 A near cancellation

$c = (22 \pm 0.5) \text{ cm}$ and $d = (21 \pm 0.5) \text{ cm}$. Find $z = c - d$ with its absolute and relative uncertainty.

Exercise 2.3 A worse one

$a = 10.00(2) \text{ cm}$, $b = 9.95(2) \text{ cm}$. Find $y = 1/(a - b)$ with its absolute and relative uncertainty.

Exercise 2.4 Powers count twice

A resistor's dissipation is computed as $P = V^2/R$ from $V = (10.0 \pm 0.2) \text{ V}$ and $R = (50 \pm 0.5) \Omega$. (a) Find P , ΔP and ε_P . (b) Which reading should you improve first?

Chapter 3

The analog oscilloscope

Exercise 3.1 Placing the spot

Where does the spot sit for $V_x = -2\text{ V}$, $V_y = 1.5\text{ V}$ with $K_x = 0.5\text{ V/div}$, $K_y = 1\text{ V/div}$?

Exercise 3.2 Choosing the scales

A signal swings between $V_{min} = -0.2\text{ V}$ and $V_{max} = 1.0\text{ V}$ at $f = 2\text{ kHz}$, and you want at least $N = 3$ periods on screen. (a) Pick the vertical scale (8 divisions, 1–2–5 sequence). (b) Pick the horizontal scale (12 divisions) and count the periods displayed. (c) What goes wrong with $100\text{ }\mu\text{s/div}$?

Exercise 3.3 Reading uncertainty: V_{pp}

$K_y = 0.5\text{ V/div}$ with $\varepsilon_{K_y} = 2\%$; the trace spans $L = 4\text{ div}$ read to $\delta L = 0.2\text{ div}$. Find V_{pp} and its relative and absolute uncertainty.

Exercise 3.4 Reading uncertainty: period and frequency

With $K_T = 10\text{ }\mu\text{s/div}$ ($\varepsilon_{K_T} = 2\%$) one period spans $L = (5.0 \pm 0.1)\text{ div}$. (a) Find T and $f = 1/T$ with their uncertainties. (b) On $20\text{ }\mu\text{s/div}$, four periods span $(10.0 \pm 0.1)\text{ div}$: redo ε_T and say why it improved.

Chapter 4

The digital oscilloscope

Exercise 4.1 Predicting an aliased trace

A DSO samples at $f_S = 1 \text{ MSa/s}$; the input is a sine at $f_{in} = 800 \text{ kHz}$. (a) What frequency does the screen show? (b) What f_S (linear interpolator, ~ 10 points/period) would display the real signal acceptably?

Exercise 4.2 Two inputs, one trace

A DSO samples at $f_S = 50 \text{ MSa/s}$. (a) What does it display for a 49 MHz sine? (b) And for a 51 MHz sine — can the two inputs be told apart? (c) What minimum f_S would you pick to view a 2 MHz sine with a linear interpolator?

Exercise 4.3 Rise time near the scope's limit

$B = 100 \text{ MHz}$ ($\pm 10\%$), so the scope's own rise time is $T_B = 0.35/B$. On $K_x = 1 \text{ ns/div}$ the edge reads $N_{div} = 4.5$ (so $T_{meas} = 4.5 \text{ ns}$). Find T_{Rise} and comment on its uncertainty.

Chapter 5

The digital multimeter

Exercise 5.1 When do you need 4 wires?

Ordinary leads, $R_c = 0.1\ \Omega$ each. (a) What is the 2-wire error for $R_x = 100\ \Omega$? (b) Below which R_x does the 2-wire error exceed 1 %?

Exercise 5.2 Square wave on a sine-calibrated meter

An average-sensing meter, calibrated for sines ($\times 1.11$), is given a 10 V peak *square* wave. What does it display, and what is the true RMS?

Exercise 5.3 Sine plus offset

$v(t) = 1.5 \sin \omega t + 2$ (in volts). Find the true RMS value.

Chapter 6

Laboratory: Bench measurements

Exercise 6.1 Sample rate from the dots

In dots mode, 60 dots span 12 divisions at 5 ns/div on CH1 alone. (a) Find the sample period T_s and rate f_s . (b) What is f_s once CH2 is also enabled?

Chapter 7

Answers

Final results only — if your method is shaky, the matching chapter of the *manual* shows how each is obtained.

Chapter 1 — Measurements and the SI system

1.1 (a) Indirect ($T = t_{10}/10$; only t_{10} carries uncertainty). (b) Direct (traceable via the thermometer's calibration). (c) Indirect ($R_{in} = V_M R / (V_S - V_M)$; depends on V_M , V_S and the known R).

1.2 (a) The DMM is *direct*; $R = V/I$ is *indirect*. (b) DMM: traceable only if its calibration is current. V/I : relative uncertainties add, $\varepsilon_R = \varepsilon_V + \varepsilon_I$.

Chapter 2 — Measurements and uncertainty

2.1 (a) 25 ± 1 cm, $\varepsilon = 4\%$. (b) 5 ± 1 cm, $\varepsilon = 20\%$. (c) 100 ± 10 cm², $\varepsilon = 10\%$. (d) 1.0 ± 0.1 , $\varepsilon = 10\%$.

2.2 $z = 1$ cm, $\Delta z = 1$ cm, $\varepsilon_z = 100\%$.

2.3 $y = 20$ /cm, $\Delta y = 16$ /cm, $\varepsilon_y = 80\%$.

2.4 (a) $P = (2.00 \pm 0.10)$ W, $\varepsilon_P = 5\%$. (b) The *voltage* (it counts twice: $\varepsilon_P = 2\varepsilon_V + \varepsilon_R$).

Chapter 3 — The analog oscilloscope

3.1 $x = -4$ div, $y = 1.5$ div (top-left quadrant).

3.2 (a) $K_y = 0.2$ V/div. (b) $K_x = 200$ μ s/div, showing ≈ 4.8 periods. (c) 100 μ s/div shows only 2.4 periods — fewer than the 3 requested.

3.3 $V_{pp} = 2.00(14)$ V, $\varepsilon_{V_{pp}} = 7\%$.

3.4 (a) $T = (50 \pm 2)$ μ s ($\varepsilon_T = 4\%$), $f = (20.0 \pm 0.8)$ kHz. (b) $\varepsilon_T = 3\%$ — spreading the reading over four periods divides the reading error by four.

Chapter 4 — The digital oscilloscope

4.1 (a) A clean 200 kHz sine (aliased). (b) $f_S \approx 8$ MSa/s.

4.2 (a) A 1 MHz trace. (b) Also 1 MHz — the two inputs are *indistinguishable*. (c) $f_S \geq 20$ MSa/s.

4.3 $T_B = 3.5 \text{ ns}$, $T_{Rise} = \sqrt{4.5^2 - 3.5^2} = 2.83 \text{ ns}$, but $\varepsilon_{T_{Rise}} \approx 32 \%$ (near-cancellation — use a faster scope).

Chapter 5 — The digital multimeter

5.1 (a) $R_{meas} = 100.2 \Omega$, error 0.2% . (b) $R_x < 20 \Omega$.

5.2 Displays 11.1 V (wrong); true RMS = 10 V .

5.3 $V_{RMS} = \sqrt{1.06^2 + 2^2} = 2.26 \text{ V}$.

Chapter 6 — Laboratory: Bench measurements

6.1 (a) $T_s = 1 \text{ ns}$, $f_s = 1 \text{ GSa/s}$. (b) $f_s = 500 \text{ MSa/s}$ (one ADC shared between channels).